

## Where are the jobs?

- ▶ EDA companies: Mentor Graphics, Synopsys, Cadence
- ▶ Semiconductor companies: STMicroelectronics, Infineon, NXP Semiconductors, Qualcomm, etc
- ▶ Electronics system companies (which do system design, PCB design): Freescale, STMicroelectronics, NXP Semiconductors
- ▶ Automotive companies: Samsung, LG Electronics, Hyundai
- ▶ Telecom companies: Huawei, ZTE, Samsung, LG Electronics
- ▶ Consumer electronics companies: Samsung, LG Electronics, Whirlpool, Voltas, Godrej

Panicker explains, "Those willing to make a career in this stream should undertake at least one front-end project or FPGA project while they are in engineering institutes. This will make them understand the VLSI flow."

But, unfortunately, there is dearth of right education and VLSI labs.

Dr Pradip Dutta, corporate vice president at Synopsys Inc. and managing director of Synopsys (India), emphasises, "There is definitely dearth of good teachers in this space. The reason is that everyone who comes in the industry wants to go and work in a multinational and nobody wants to teach. They find better growth prospects in working in multinational companies rather than teaching in colleges. Also, there is lack of good VLSI labs, but the situation is improving with time."

Thankfully, the industry is coming forward to provide assistance. Panicker shares, "The industry is providing students the right kind of tools and setting up labs so that they can have a hands-on experience of the EDA tools. For example, STMicroelectronics came up with the idea to improve the situation in the engineering colleges that we hire from. We jointly approached IIT Delhi and told them that we want to run certain programmes for the seventh-semester and eighth-

semester students. We started running programmes and courses for these students, which enabled students to do projects. At this stage in the engineering courses, students are expected to do projects which are industry-worthy. This makes the students ready with minimum requirement of understanding the processes required to make them employable in companies like ours."

### Get industry-ready before getting on the job

The syllabus in your college is not enough to fetch you a job in the EDA industry. You need to have an extra bit to get selected. There are training institutes like EFY Tech Center which provide hands-on training in VLSI courses, making you industry-ready.

Dutta clarifies, "Finishing schools do have a role in the lives of EDA engineers, the reason being many of the graduates from the institutes are not ready for the industry. For those who want to specialise in a particular area like verification of circuits or work in digital implementation, there are some high-end finishing schools in the VLSI sector. They offer a six-month to one-year course, which helps the candidates become industry-ready and make a career in EDA."

Panicker adds, "Students after finishing their courses go to these training institutes and learn the semiconductor processes. These are the likes of NIIT and Aptech that are teaching the software part of it. So even though the engineers learn computer engineering in the engineering colleges, they still undergo some training from these institutions. Similarly, in our space, there are companies like RV VLSI and Sandeepani. Most engineers undergo a diploma programme of 3-4 months after their degree, to get equipped in the VLSI and semiconductor flow. They are then ready to be hired by VLSI companies like Wipro and Qualcomm."

Some companies too put the freshers on a learning plan. Cadence offers on-job training for three to six months.

Pathak says, "EDA is a very different segment altogether so we definitely need to acclimatise people and align them with what EDA is all about. That takes a little bit of its own time. With time, they get into the normal cycle. So that's how we move ahead in terms of our fresher hiring. And then there is lateral hiring, where companies reach out to the market and hire people for specific skillsets and certain job levels which include mid-level and senior-level positions as well."

### The road ahead

While hiring, the EDA industry looks for brains. Pathak explains: "EDA is a very niche segment. So the hiring is very qualitative. We surely do not hire in thousands like any other IT company but we pick the crème de talent. We cannot and do not hire anyone and everyone. The candidate has to be the



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